

GitHub Actions

What is GitHub actions?

- It is a continuous integration and continuous delivery (CI/CD) platform
- It allows you to automate build, test, and deployment pipeline
- I can create workflows that build and test every pull request to your repository, or deploy merged pull requests to production.
- It provides Linux, Windows, and macOS virtual machines to run your workflows, or you can host your own self-hosted runners in your own data centre or cloud infrastructure.

Components of GitHub Actions:

1.workflow

- A workflow is a configurable automated process that will run one or more jobs.
- Workflows are defined by a YAML file checked in to your repository and will run when triggered by an event in your repository, or they can be triggered manually, or at a defined schedule.
- Workflows are defined in the .github/workflows directory in a repository.
- A repository can have multiple workflows
 - Building and testing pull requests
 - Deploying your application every time a release is created
 - Adding a label whenever a new issue is opened

2.Events

- An event is a specific activity in a repository that triggers a workflow run.

For example:

an activity can originate from GitHub when someone creates a pull request, opens an issue, or pushes a commit to a repository.

---You can also trigger a workflow to run on a [schedule](#), by [posting to a REST API](#), or manually.

3. Jobs

- A job is a set of steps in a workflow that is executed on the same runner.
- Each step is either a shell script that will be executed, or an action that will be run.
- Steps are executed in order and are dependent on each other.
- Since each step is executed on the same runner, you can share data from one step to another.
 - For example:
 - you can have a step that builds your application followed by a step that tests the application that was built

4. Actions

- An action is a pre-defined, reusable set of jobs or code that performs specific tasks within a workflow, reducing the amount of repetitive code you write in your workflow files.

Actions can perform tasks such as:

- Pulling your Git repository from GitHub
- Setting up the correct toolchain for your build environment
- Setting up authentication to your cloud provider

- You can write your own actions, or you can find actions to use in your workflows in the GitHub Marketplace.

5. Steps

- Steps are individual tasks executed in sequence within a specific job.
- A step can either run a shell script using the run keyword or execute a reusable "action" using the uses keyword

6. Runners

- A runner in GitHub Actions is a server that executes the jobs within a workflow.
- It is the machine where your build, test, and deployment processes actually run, essentially acting as the compute environment for your CI/CD pipeline